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Web-Retrieval Fact-Checking is a process about evaluating the truthfulness of an information, in this case about Health claims. The current problem is the following : health claims today are used and accessible in a broad way, without being properly verified. This causes a drastic issue as the health sector is not one that can allow a mistake without consequences, that can vary from being inoffensive to lethal. These problems can be even more frequent now that we entered an era of information by AI, that can hallucinate things, thus provoking accidents.

Research were already done in this domain as shown with the project PubhealthTab, that tried to use the PUBHEALTH dataset as a source to make a filtered dataset of fact-checked informations: https://aclanthology.org/2022.findings-naacl.1/

For this project, the goal will be to compare different approaches by using different ranking systems that rank the veracity of a number of claims. The question we are researching is "What would be the best ranking system of health claims fact-checking to enable a maximum level of prevention ?"

We will try to focus our results on a specific region of these claims, as they are sparse and not all interesting. For example, it would be interesting to filter only claims related to medicine and risk associated.

In order to do so, we will sample health claims from the dataset across the different veracity labels (true, false, unproven, mixture) and use them for our experiment.
Then using a web search API (Tavily / Serper), we will retrieve the top 10 websites for each claim blacklisting the one used by the original research of the PUBHEALTH project to avoid leaking answers. We'll then chunk these pages into a few hundred words.
For each claim, we will rank candidate passages from the previously selected pool against the claim in order to retrieve the top-k passages per method
Here are the methods we'll rank alongside the baseline API of the web search :

1. BM25
2. Dense bi-encoder retrieval
3. Cross-encoder reranking

We will feed the top-k passage of each method into a local LLM (Qwen / Mistral / Gemma …) predicting veracity labels and evaluating verifiability rate. Prior to that, we will run the verifier without evidence to test its knowledge.
We hypothesise that due to its lexical nature, BM25 will underperform compared to the others that would look into the semantic side.
Finally, we'll manually categorize mismatches into retrieval errors, verifier reasoning failures, and temporally outdated web evidence.
In order to do so, we will use the PUBHEALTH benchmark dataset (Kotonya & Toni, 2020), available via Hugging Face or on kaggle:
https://www.kaggle.com/datasets/ersindemirel/pubhealthdataset